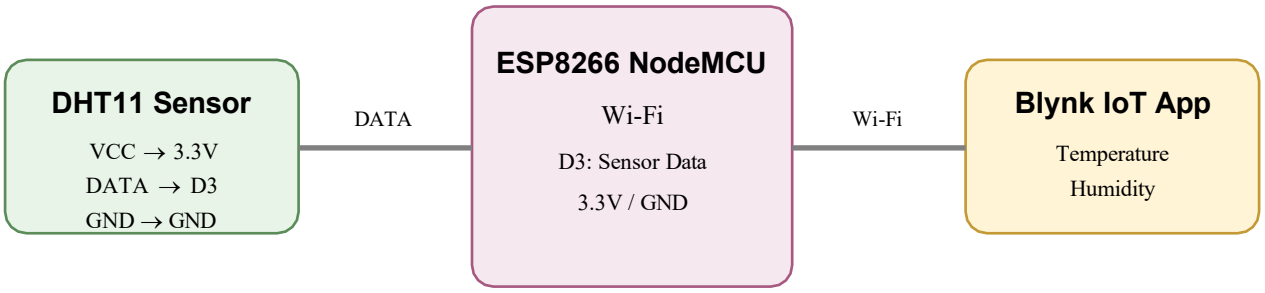
ASSIGNMENT - 2

Aim: Read **Temperature & Humidity** using ESP8266 and DHT11 Sensor and display the sensor data on the Blynk IoT App.

Circuit Diagram:



Hardware Required

- ESP8266 NodeMCU
- DHT11 (or DHT22) Sensor
- Breadboard
- Jumper wires
- USB Cable
- Wi-Fi connection
- Blynk IoT Account

Procedure

1. Connect the DHT11 sensor: VCC to 3.3V, DATA to D3, and GND to GND.
2. Create a Blynk IoT template named Temperature Monitoring and select ESP8266 with Wi-Fi.
3. Create Temperature (V0) and Humidity (V1) datastreams.
4. Create the device, obtain the Auth Token, and configure the mobile dashboard with Gauge widgets.
5. Upload the program to the ESP8266 and observe the sensor readings.

Code

```
#include "DHT.h"
#define DHTTYPE DHT11
#define dht_dpin D3
DHT dht(dht_dpin, DHTTYPE);

void setup() {
  dht.begin();
  Serial.begin(9600);
  Serial.println("Humidity and
  temperature\n\n"); delay(700);
}

void loop() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
  Serial.print("Current humidity = ");
  Serial.print(h);
  Serial.print("%  ");
  Serial.print("temperature =
  "); Serial.print(t);
  Serial.println("C  ");
  delay(800);
}
```

Output:

